1.36 Integration: Indefinite Integrals

Chapter 13 · Original slide 3 · Integrating {x}^{n} terms

Integration is the opposite of differentiation.

(For this reason it is also called '*antidifferentiation*')

Gradient Function

Function

multiply by power

reduce power by 1

$15x^2$

$5x^3$

divide by new power

increase power by 1

However, there's one added complication...

Find y when $\frac{dy}{dx} = 3x^2$

c is known as a constant of integration

Chapter 13 · Original slide 4 · Examples

Find y when:

You should always include the c.

$\frac{dy}{dx} = 4x^3 \qquad y =$

$\frac{dy}{dx} = x^5 \qquad y =$

You could also write as $\frac{x^6}{6}$

$\frac{dy}{dx} = 3x^{1/2} \qquad y =$

When you divide by a fraction, you multiply by the reciprocal.

Textbook Error on Pg289: "When integrating polynomials", but Example 3 is not a polynomial because it has negative and fractional powers.

1.36 Integration: Indefinite Integrals

Chapter 13 · Original slide 5 · More Fractional Examples

Find y when:

$\frac{dy}{dx} = \frac{4}{\sqrt{x}}$ $y =$

$\frac{dy}{dx} = 4x^{2/3}$ $y =$

When we divide by $\frac{1}{2}$ we multiply by the reciprocal, i.e. 2.

$\frac{dy}{dx} = 5x^{-2}$ $y =$

$\frac{dy}{dx} = 10x^{-2/7}$ $y =$

Chapter 13 · Original slide 6 · Test Your Understanding

Find $f(x)$ when:

$\mathrm{f}'(x) = x^2 - 1$ $\mathrm{f}(x) =$

$\mathrm{f}'(x) = \sqrt[3]{x}$ $\mathrm{f}(x) =$

Note: In case you're wondering what happens if $\frac{dy}{dx} = \frac{1}{x} = x^{-1}$, the problem is that after adding 1 to the power, we'd be dividing by 0. You will learn how to integrate $\frac{1}{x}$ in Year 2.

$\mathrm{f}'(x) = 2x + 7$ $\mathrm{f}(x) =$

$\mathrm{f}'(x) = \frac{2}{x^7}$ $\mathrm{f}(x) =$

$\mathrm{f}'(x) = 33x^{5/6}$ $\mathrm{f}(x) =$

1.36 Integration: Indefinite Integrals

Chapter 13 · Original slide 7 · Exercise 13A – Y1 pages 289-290

4. Find y given that $\frac{dy}{dx} = (2x + 3)^2$

5. Find $f(x)$ given that
$f'(x) = 3x^{-2} + 6x^{\frac{1}{2}} + x - 4$

Chapter 13 · Original slide 8 · Exercise 13A – Y1 pages 290

Challenge. Find y given that
$\frac{dy}{dx} = \left(2\sqrt{x} - x^2\right)\left(\frac{3+x}{x^5}\right)$

1.36 Integration: Indefinite Integrals

Chapter 13 · Original slide 9 · Integration notation

The following notation could be used to differentiate an expression:

$\frac{d}{dx}\left(5x^2\right) = 10x$

There is similarly notation for integrating an expression:

$\int 10x dx = 5x^2 + c$

The dx here means differentiating "with respect to x".

"Integrate..."

"...this expression"

"...with respect to x"

(the dx is needed just as it was needed in the differentiation notation at the top of this slide)

This is known as indefinite integration, in contrast to definite integration, which we'll see later in the chapter.

It is called 'indefinite' because the exact expression is unknown (due to the $+c$).

Advanced Note: For the moment you should think of the dx just as part of the $\int(\dots)dx$ notation, informing us what variable we're integrating with respect to. But strictly the dx actually beh

Chapter 13 · Original slide 10 · Examples

Note: The brackets are required if there's multiple terms.

Find $\int(6t^2 - 1)dt$

Note the dt instead of dx.

Find $\int(px^3 + q)dx$ where p and q are constants.

Find $\int(x^{-\frac{3}{2}} + 2)dx$

1.36 Integration: Indefinite Integrals

Chapter 13 · Original slide 11 · Test Your Understanding

Given that $y = 2x^5 + \dfrac{6}{\sqrt{x}}$, $x > 0$, find in their simplest form

(b) $\displaystyle\int y \, \mathrm{d}x$ (3)

Chapter 13 · Original slide 12 · Exercise 13B – Y1 pages 291-293

10 Given that $\mathrm{f}(x) = \dfrac{6}{x^2} + 4\sqrt{x} - 3x + 2$, $x > 0$, find $\displaystyle\int \mathrm{f}(x) \, \mathrm{d}x$.

11 Find $\displaystyle\int \left(8x^3 + 6x - \frac{3}{\sqrt{x}}\right) \mathrm{d}x$, giving each term in its simplest form.

1.36 Integration: Indefinite Integrals

Chapter 13 · Original slide 13 · Exercise 13B – Y1 pages 291-293

12 a Show that $(2+5\sqrt{x})^2$ can be written as $4+k\sqrt{x}+25x$, where k is a constant to be found.

b Hence find $\int (2+5\sqrt{x})^2 \, \mathrm{d}x$.

13 Given that $y = 3x^5 - \dfrac{4}{\sqrt{x}}$, $x > 0$, find $\int y \, \mathrm{d}x$ in its simplest form.

Chapter 13 · Original slide 14 · Exercise 13B – Y1 pages 291-293

14 $\int \left(\dfrac{p}{2x^2} + pq\right) \mathrm{d}x = \dfrac{2}{x} + 10x + c$

Find the value of p and the value of q.

15 $\mathrm{f}(x) = (2-x)^{10}$.

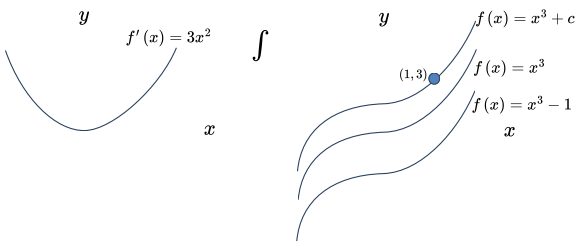
Given that x is small, and so terms in x^3 and higher powers of x can be ignored:

a find an approximation for $\mathrm{f}(x)$ in the form $A + Bx + Cx^2$

b find an approximation for $\int \mathrm{f}(x) \, \mathrm{d}x$.

1.36 Integration: Indefinite Integrals

Chapter 13 · Original slide 15 · Finding constant of integration



?

Recall that when we integrate, we get a constant of integration, which could be any real value. This means we don't know what the exact original function was.

But if we know one point on the curve, it leaves only one possibility.

Chapter 13 · Original slide 16 · Test Your Understanding

A curve with equation $y = \mathrm{f}(x)$ passes through the point $(4, 25)$.

Given that
$\mathrm{f}'(x) = \frac{3}{8}x^2 - 10x^{-1/2} + 1, \qquad x > 0$

(a) find $\mathrm{f}(x)$, simplifying each term. (5)

(b) Find an equation of the normal to the curve at the point $(4, 25)$.

Give your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found. (5)

(b)

1.36 Integration: Indefinite Integrals

Chapter 13 · Original slide 17 · Exercise 13C – Y1 pages 294-295

4 The curve with equation $y = \mathrm{f}(x)$ passes through the point $(-1, 0)$. Given that $\mathrm{f}'(x) = 9x^2 + 4x - 3$, find $\mathrm{f}(x)$.

5 $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 3x^{-1/2} - 2x\sqrt{x}$, $x > 0$.
Given that $y = 10$ at $x = 4$, find y in terms of x, giving each term in its simplest form.

Chapter 13 · Original slide 18 · Exercise 13C – Y1 pages 294-295

6 Given that $\dfrac{6x + 5x^{3/2}}{\sqrt{x}}$ can be written in the form $6x^p + 5x^q$,

a write down the value of p and the value of q.

Given that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{6x + 5x^{3/2}}{\sqrt{x}}$ and that $y = 100$ when $x = 9$,

b find y in terms of x, simplifying the coefficient of each term.